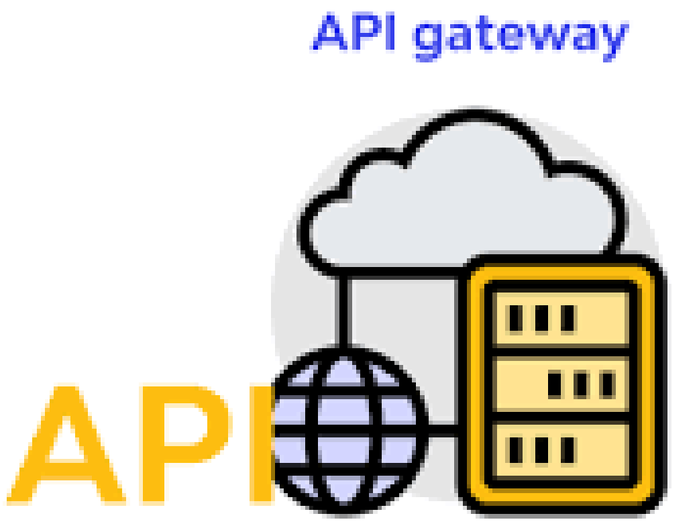
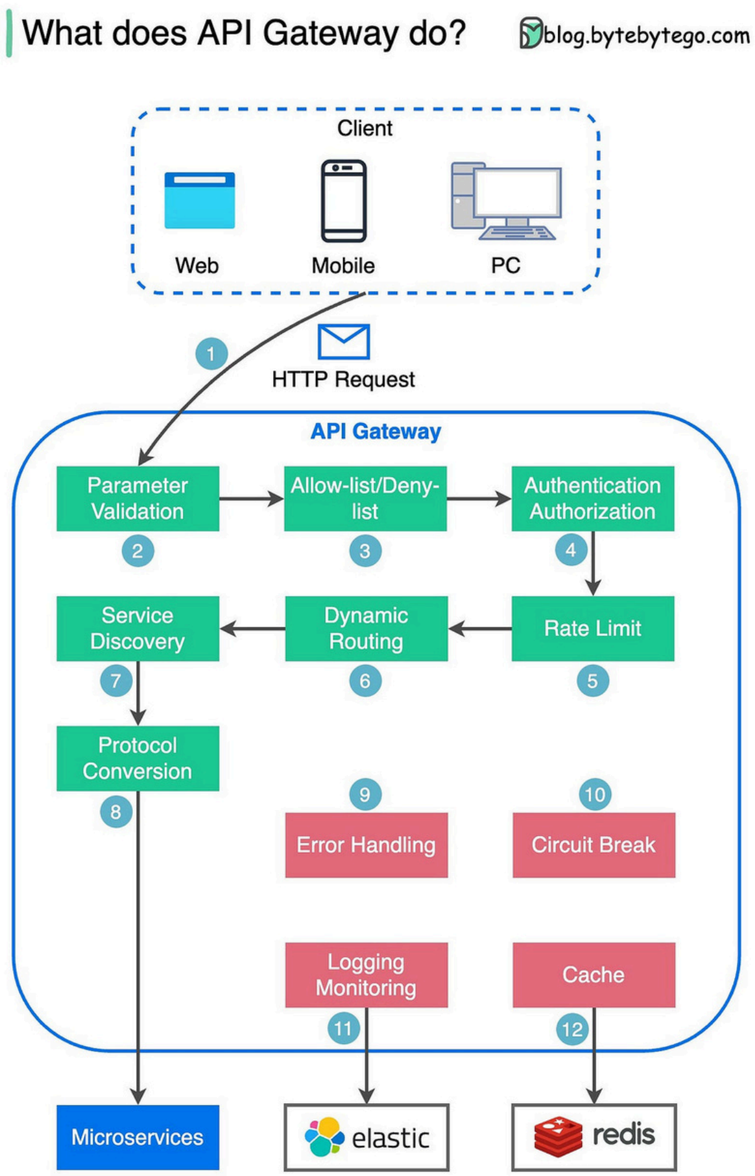
API GATEWAY

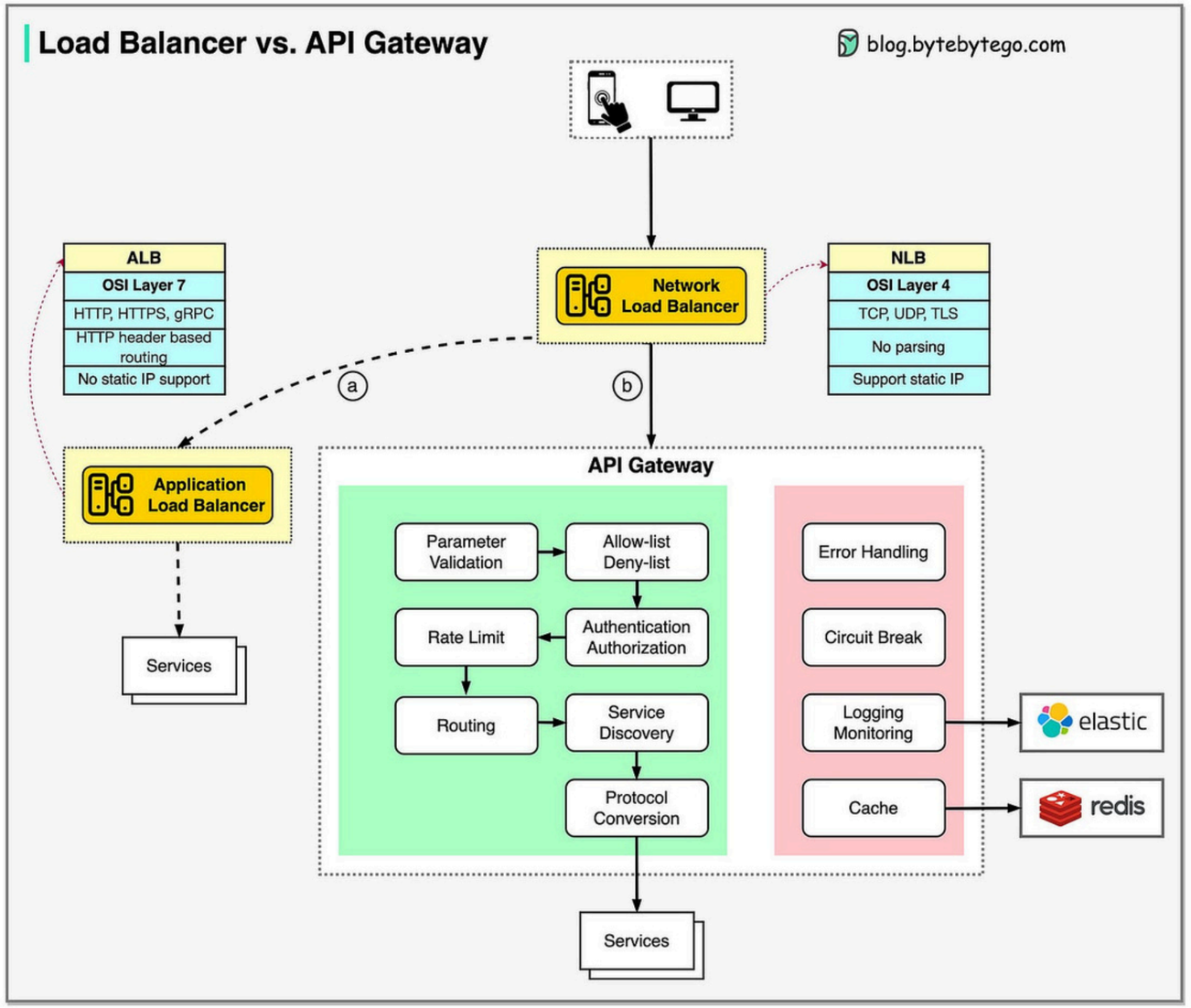


API Gateway Nedir?

- API Gateway, istemciler ile backend servisleri arasında çalışan ve tüm API trafiğini yöneten merkezi bir katmandır.
- Microservice mimarisinde her servis ayrı çalışır. Client'ın her servise ayrı bağlanması yerine gateway kullanılır.
- Cloud-native mimarilerde API gateway, güvenlik ve trafik kontrolünün merkezi noktası olarak kabul edilir.



API Gateway Nasıl Çalışır?

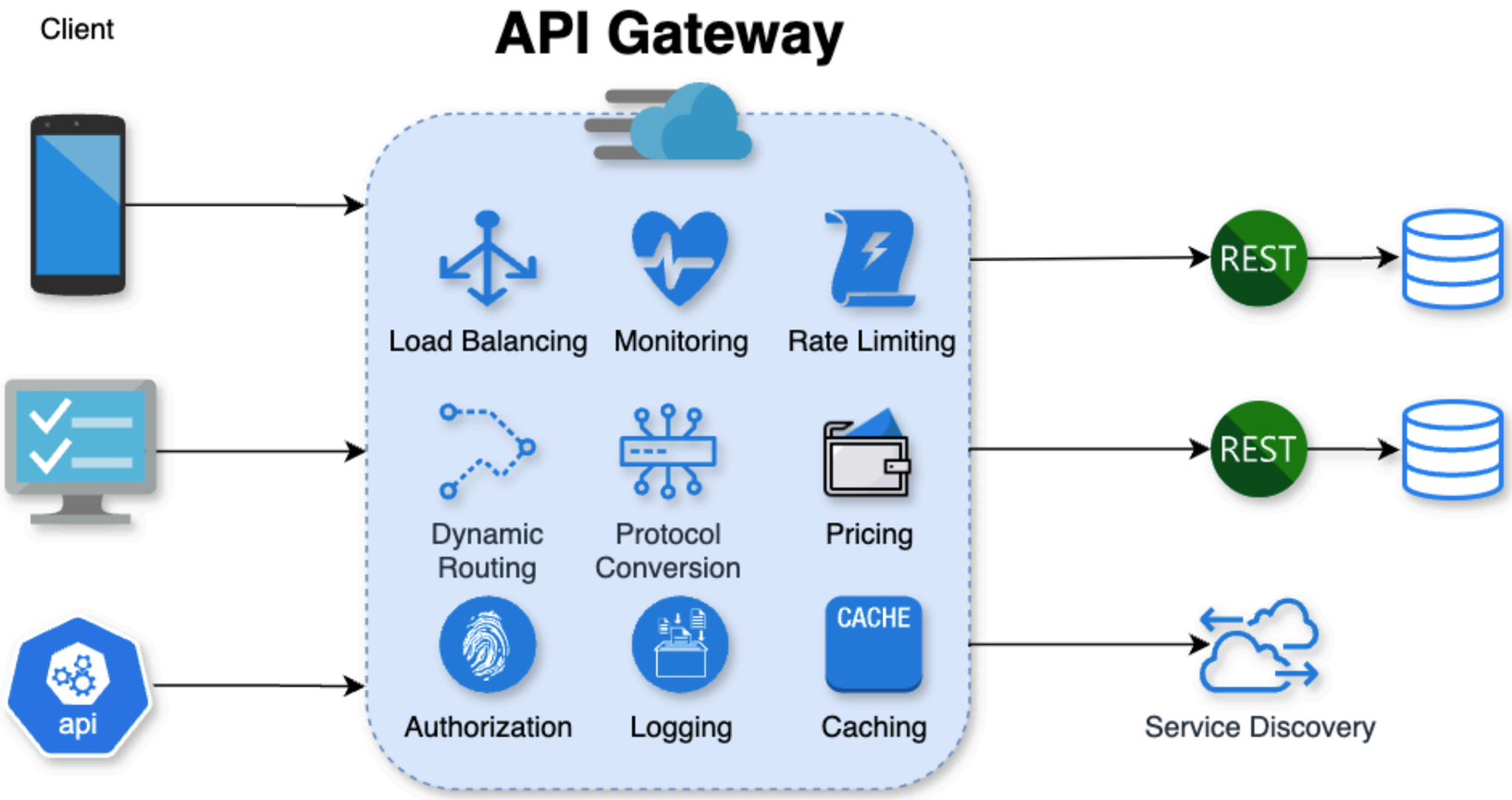


API Gateway Neden Kullanılır?

- Merkezi Güvenlik
- Observability
- Performans
- Basit Client Tasarımı

Dezavantajları

- Tek hata noktası olabilir.
- Yanlış tasarım bottleneck yaratabilir.
- Fazla logic koyulursa monolith'e dönebilir.



API Gateway Servisleri

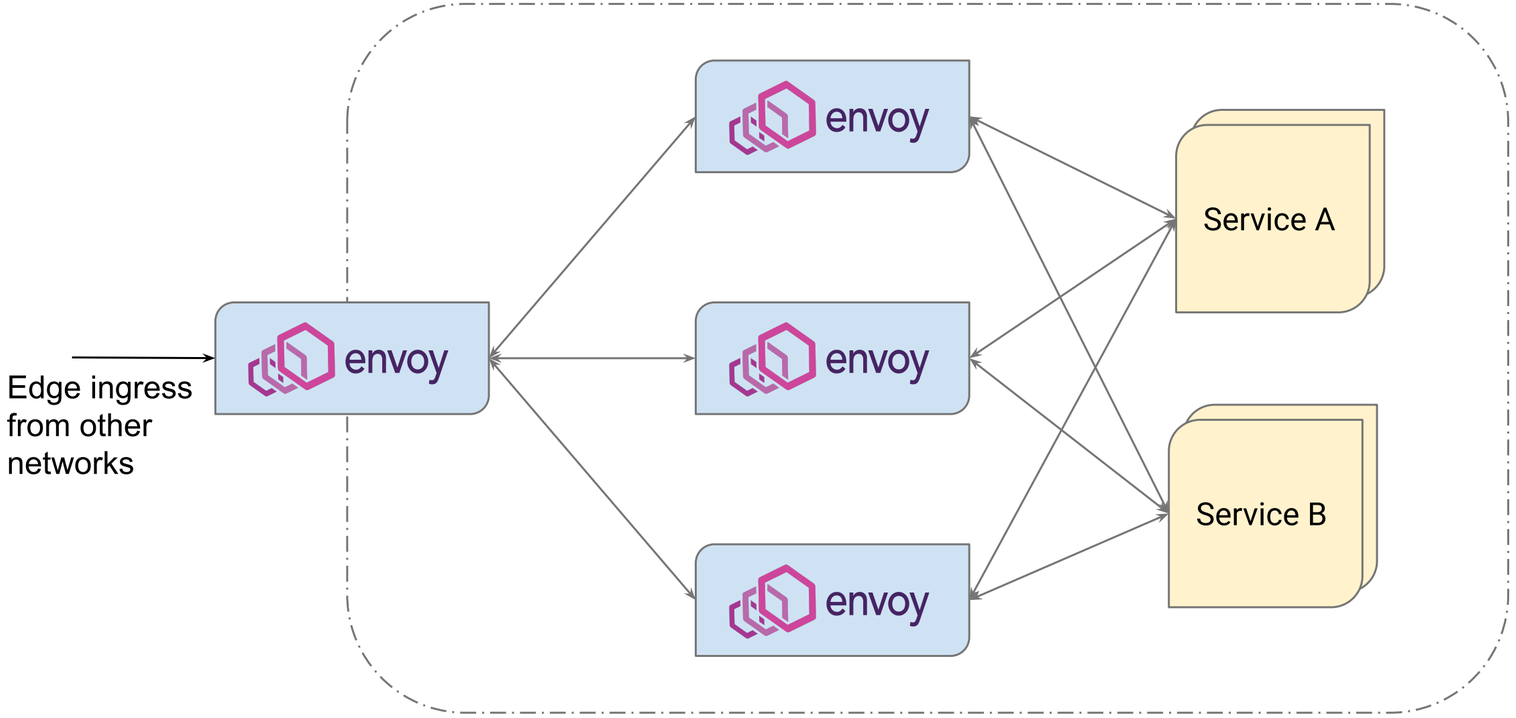
1.Envoy API Gateway

Envoy aslında tam bir API platformu değil, L7 proxy olarak çalışan bir data-plane bileşenidir.

-Gelen HTTP/gRPC trafiğini alır.

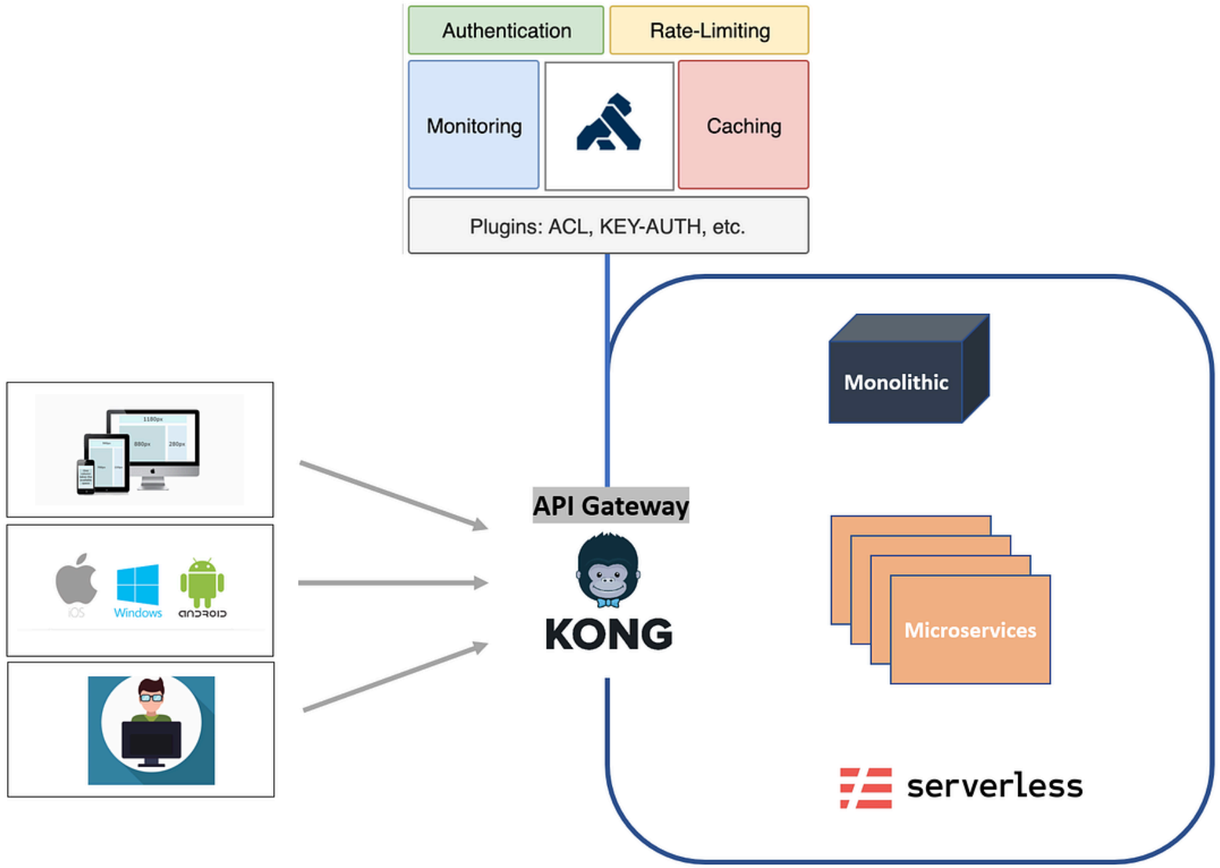
-Route kurallarına göre yönlendirir.

-Filtre zinciri (filter chain) ile auth, rate limit, telemetry uygular.



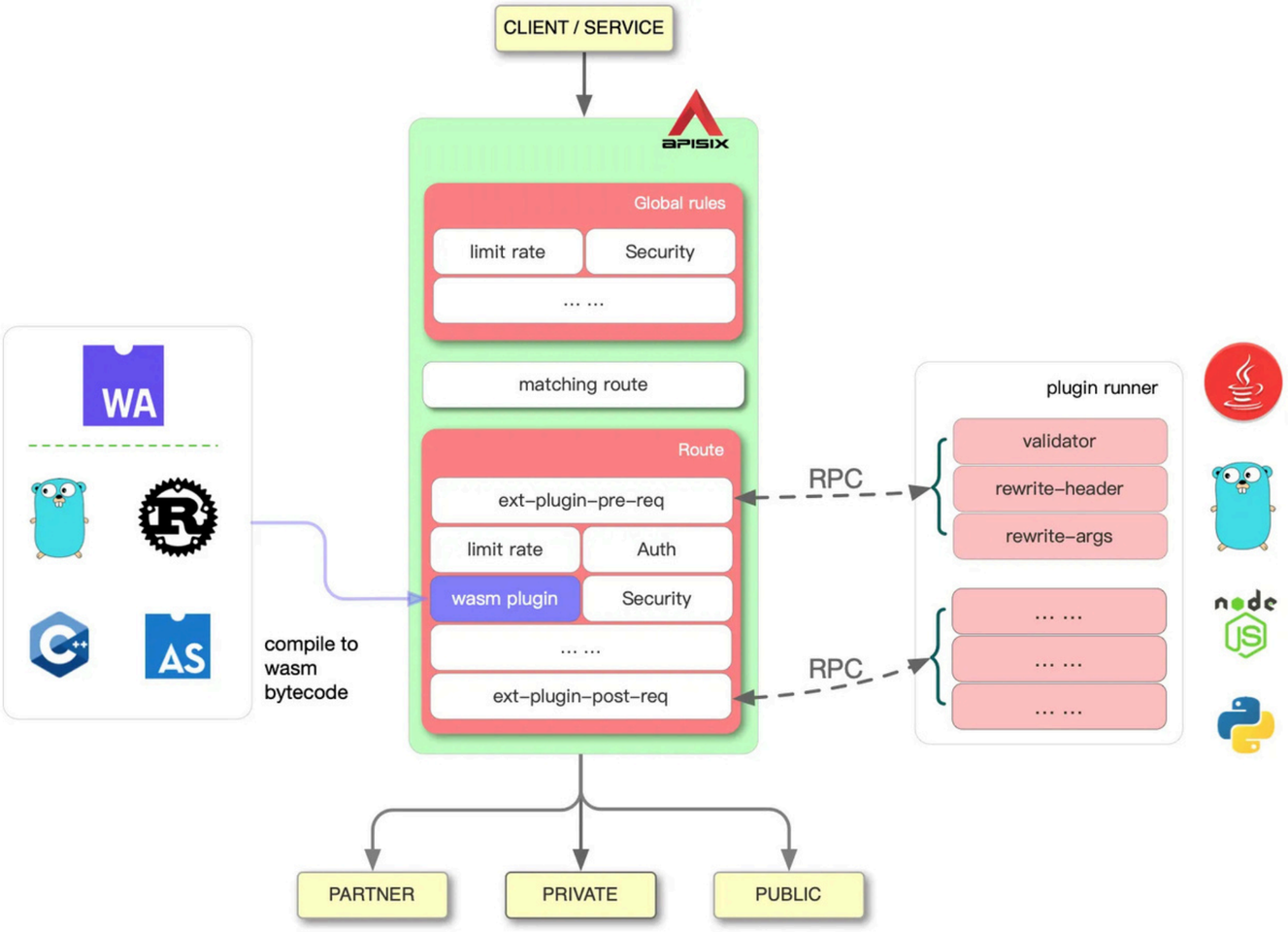
2. Kong API Gateway

Kong’un temelinde NGINX vardır. Üzerine Lua tabanlı plugin sistemi eklenmiştir.



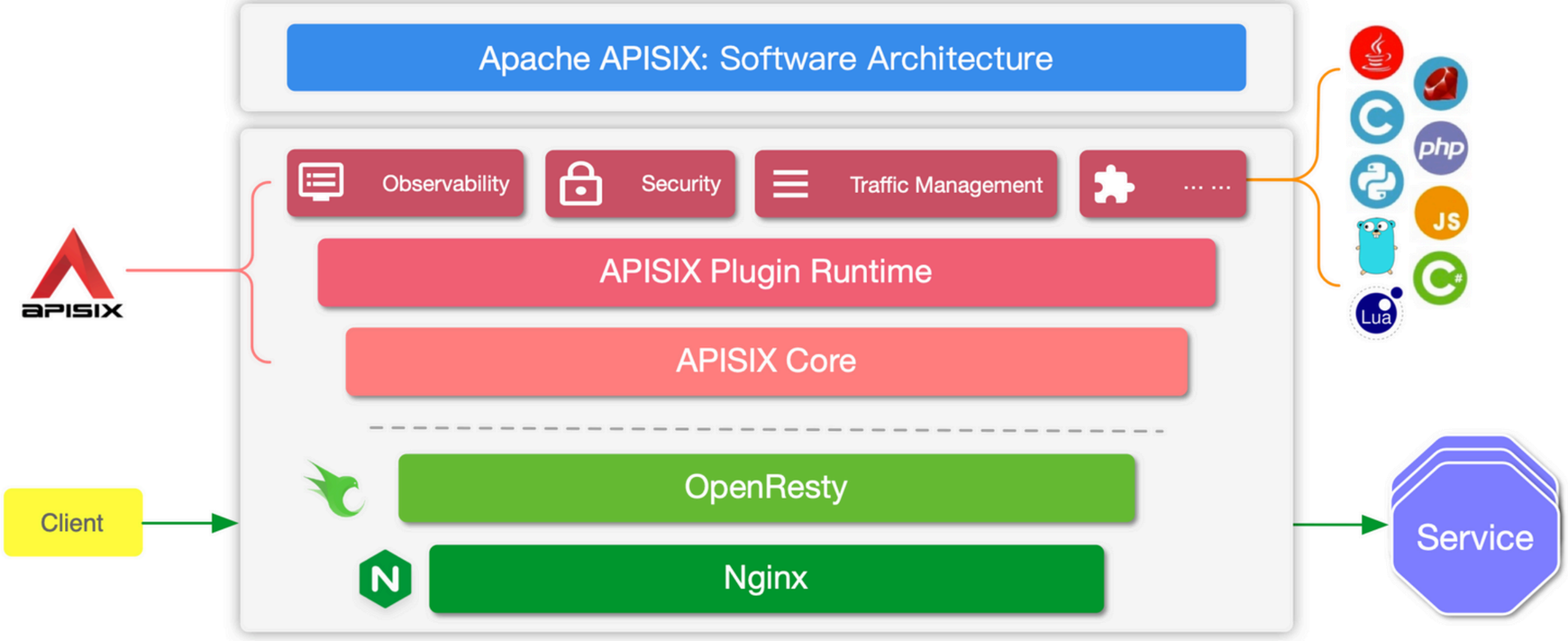
3. Apache APISIX

APISIX de NGINX tabanlıdır ama dynamic configuration kullanır.
Config değişince restart gerekmez
etcd üzerinden runtime config alır



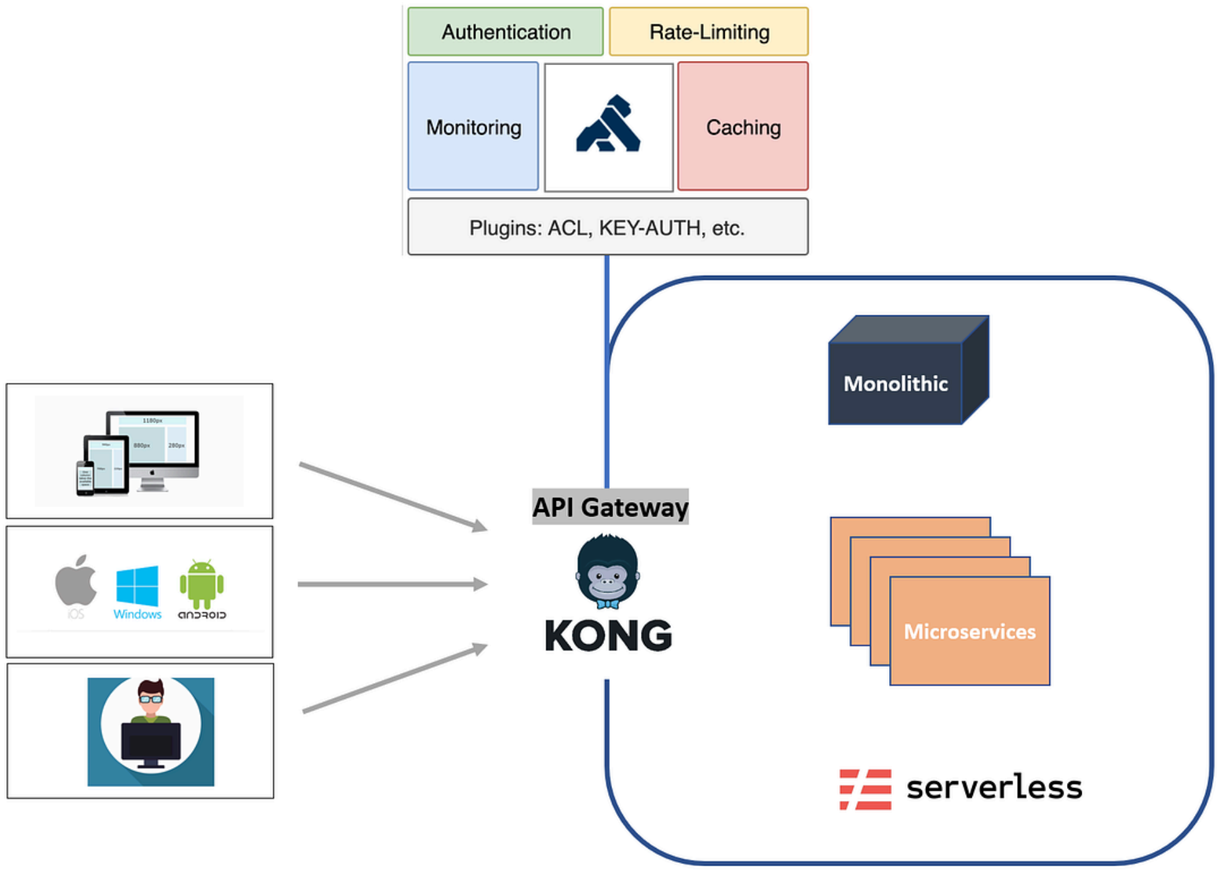
4. KrakenD

KrakenD tamamen stateless çalışır.
En önemli özelliği:
-API Aggregation
Tek request → birden fazla backend çağrısı → tek response



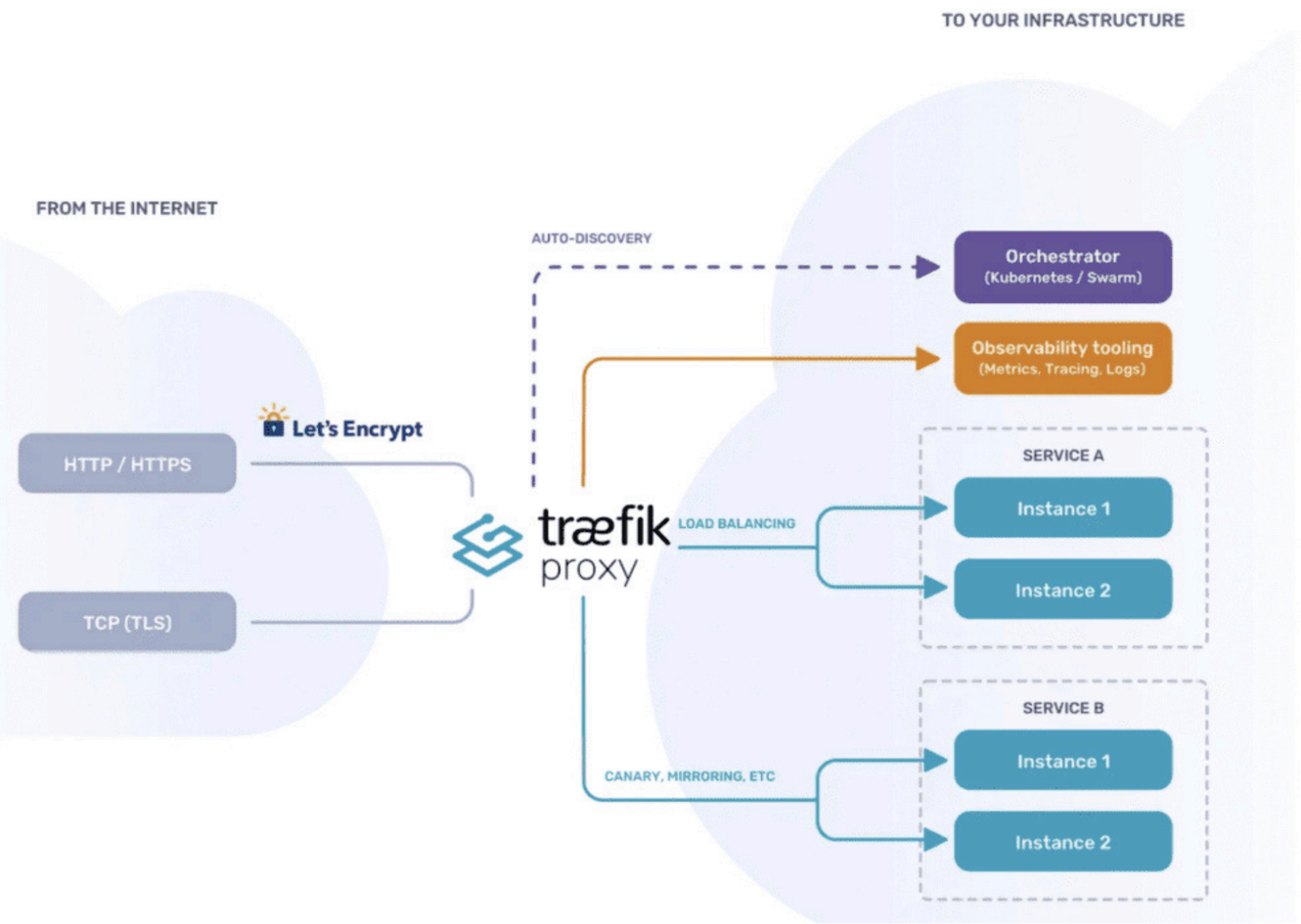
5. Tyk API Gateway

Tyk bir API lifecycle platformudur.
Gateway + Dashboard + Analytics içerir.



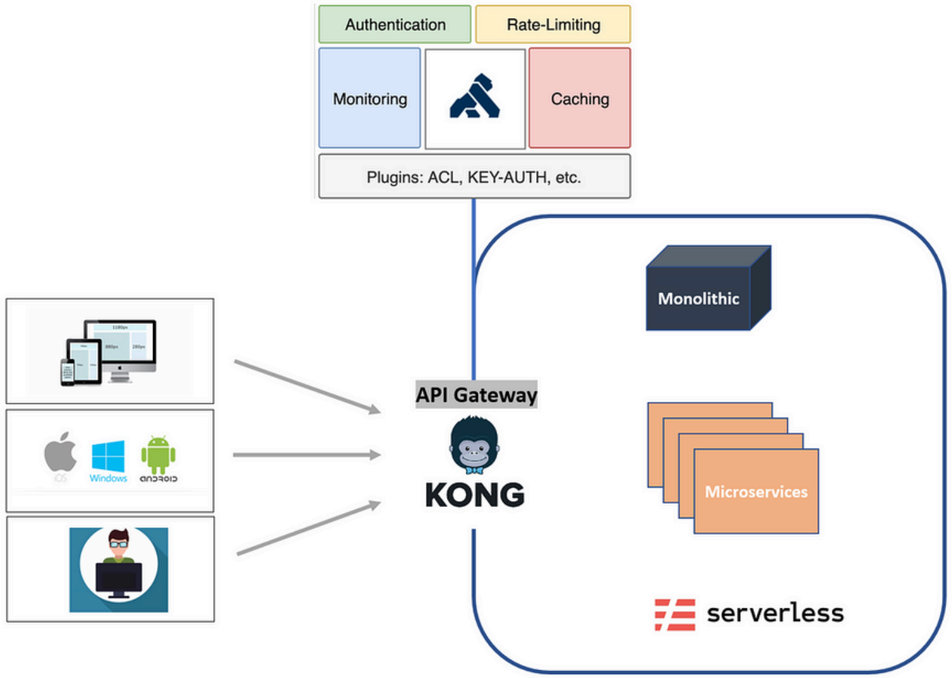
6. Traefik

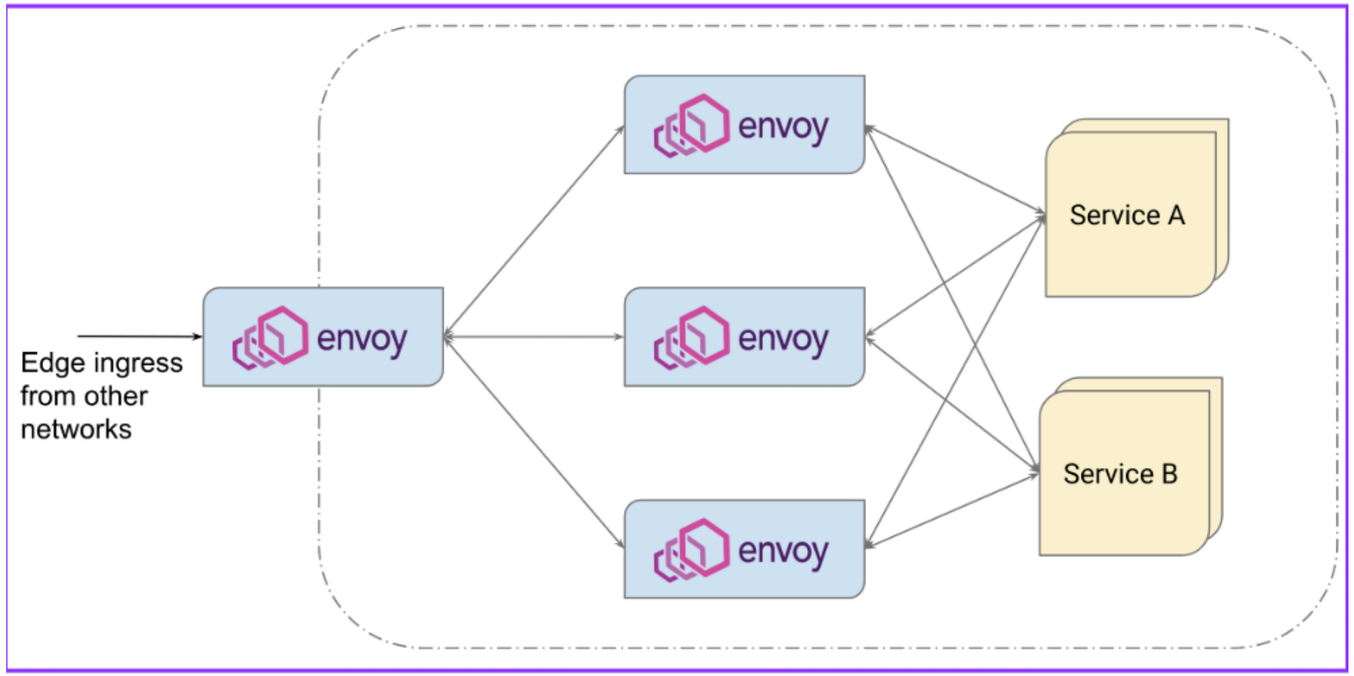
Traefik event-driven çalışır.
Docker/Kubernetes’te yeni servis açıldığında otomatik route oluşturur.



7. NGINX API Gateway

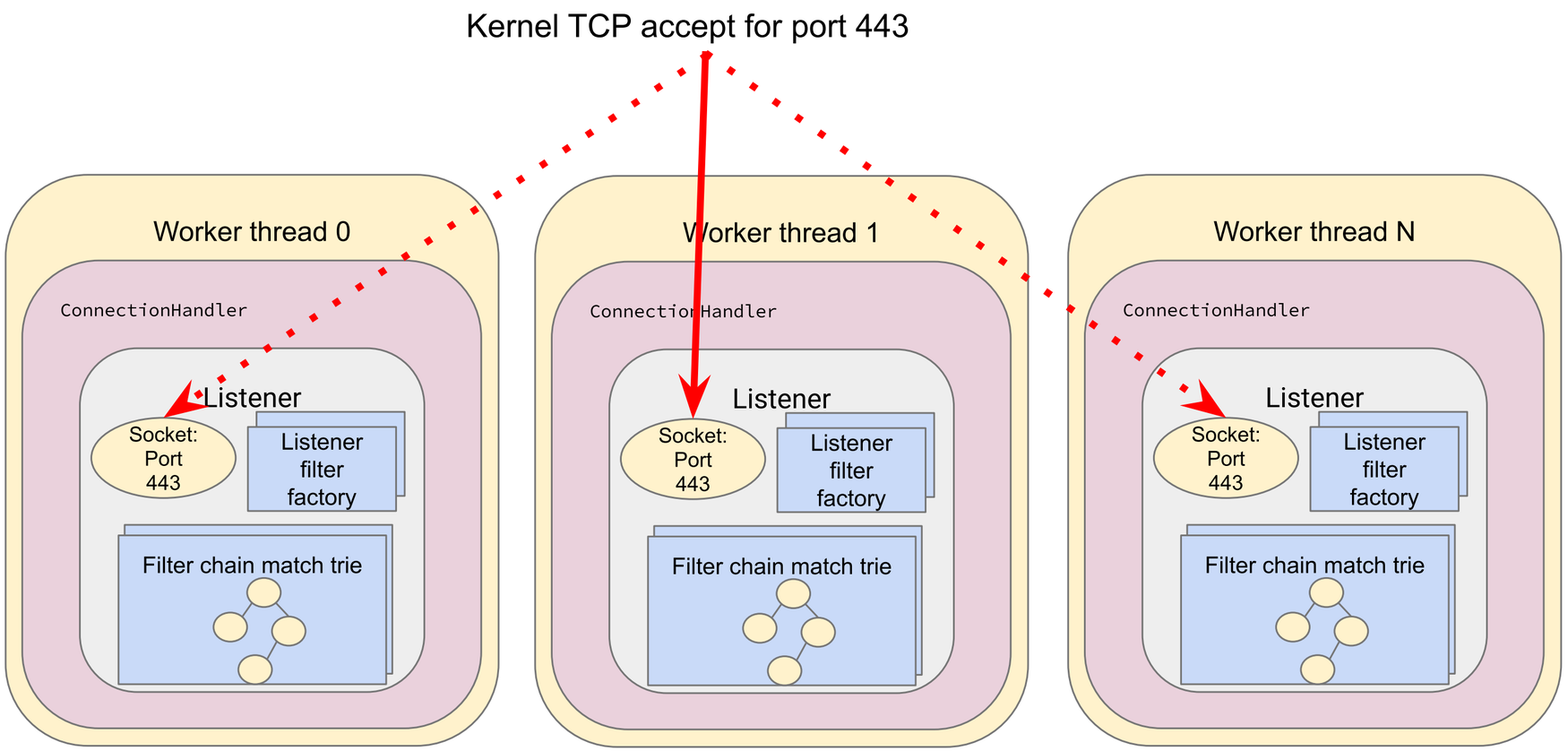
NGINX reverse proxy mantığıyla çalışır.
-Worker process modeli
-Event-driven I/O





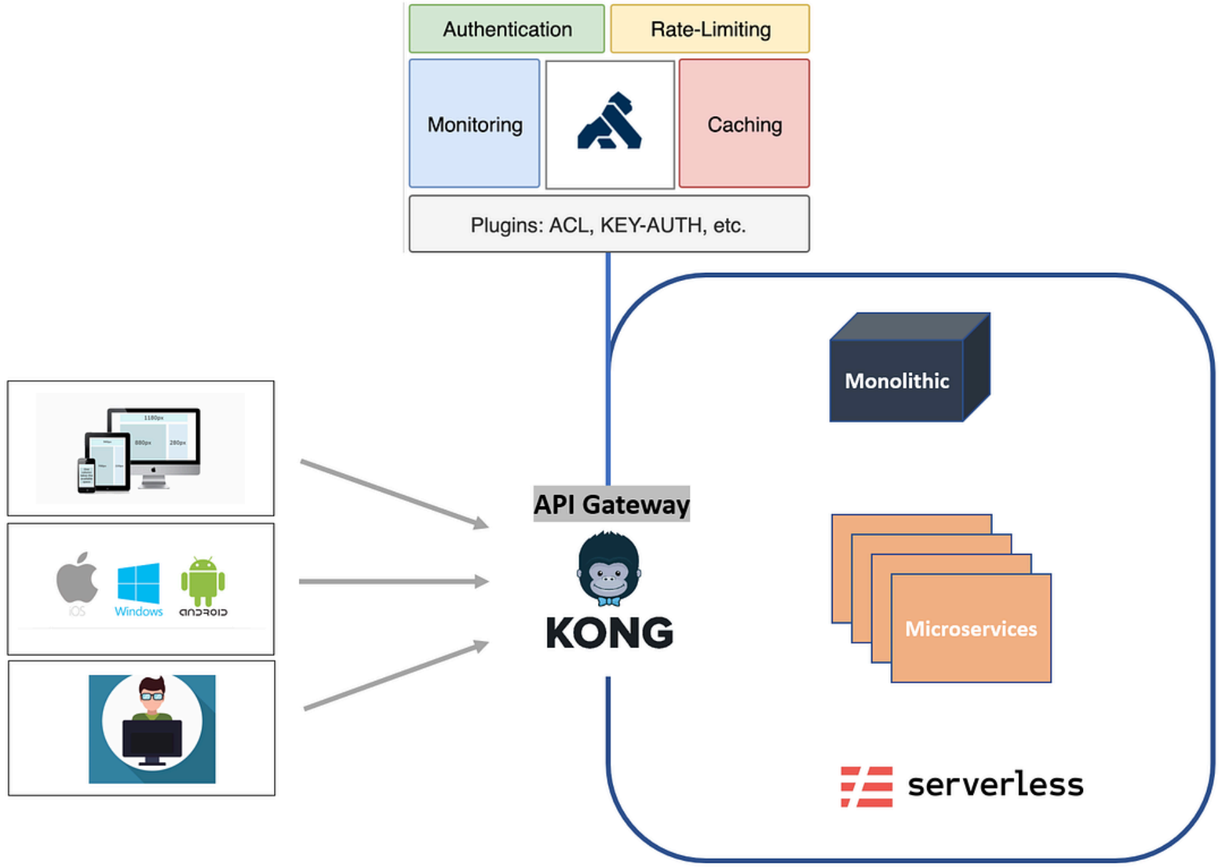
8. Ambassador / Emissary

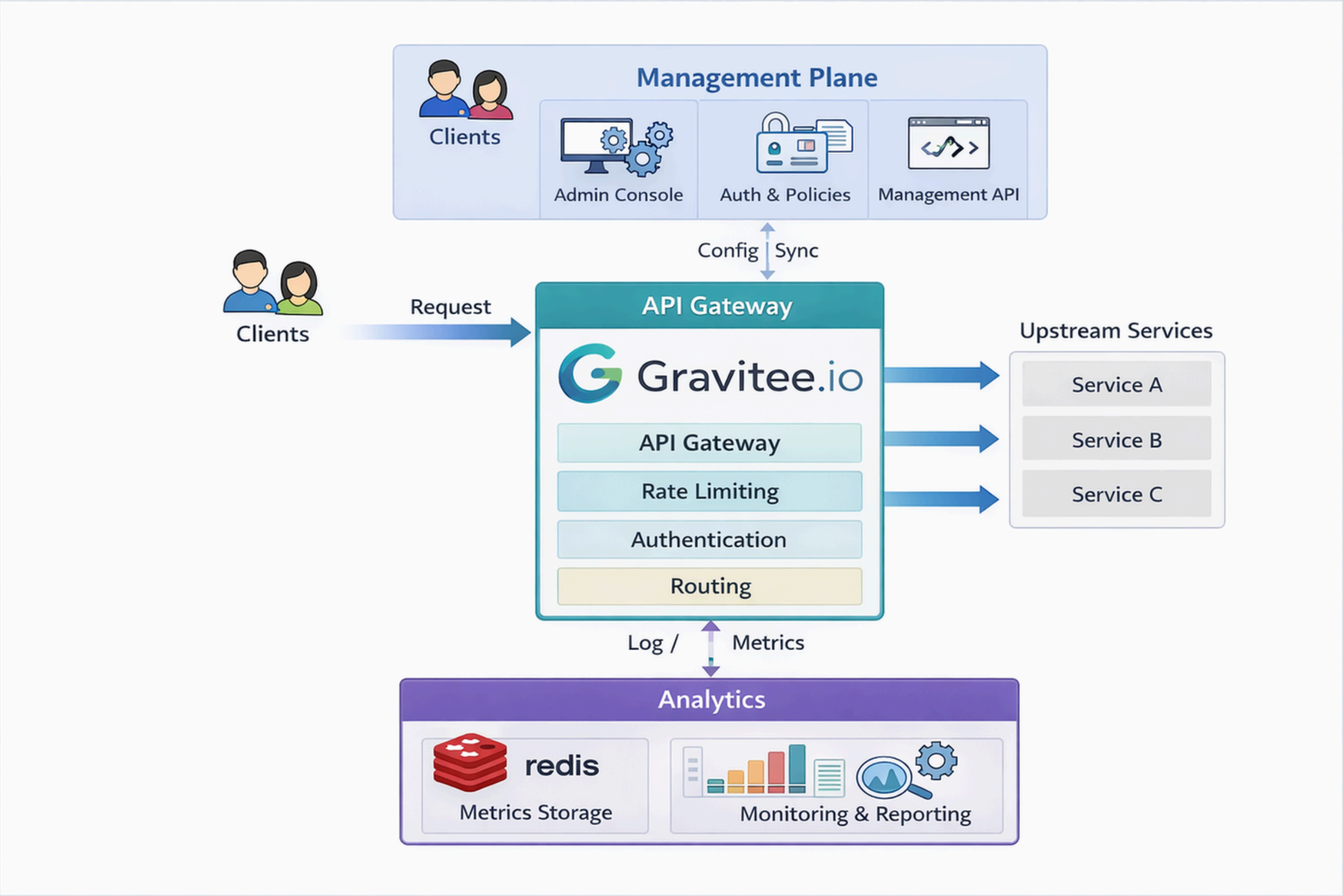
Envoy üzerine kurulu control plane sağlar.
Kubernetes CRD kullanır



9. Gloo Edge

-Envoy üzerine ek control layer’dır.
-API discovery
-GraphQL federation
-Serverless routing



10. Gravitee.io

Event-driven gateway.
Streaming API’ler için optimize edilmiştir.